

SEAM

Sign-language Emotion-Aware Multimodal Framework

Factorized Non-Manual Modeling for Emotion-Aware
American Sign Language Translation and Expressive 3D Avatar Generation

Project Overview Document

September 2026

Project Classification

Type: Research Project with Applied Components

SEAM is primarily a research project advancing the field of sign language understanding and affective computing. It combines novel machine learning research (non-manual signal factorization) with practical engineering components (real-time deployment, 3D avatar generation). The project aims to produce publishable research contributions while delivering a functional prototype.

Project Idea

Problem Statement

Sign language translation systems overwhelmingly model the manual channel (hands, arms) and discard the non-manual channel (face, brows, mouth, gaze, head, torso). This is not a cosmetic omission: in ASL the non-manual channel is *grammatically obligatory* — raised brows mark yes/no questions, a furrowed brow marks wh-questions, a head-shake marks negation. The same channel simultaneously carries affect. Existing systems therefore fail twice: they drop grammar, and they drop emotion.

Proposed Solution

SEAM addresses this with a **factorized non-manual encoder** that splits the non-manual signal into two explicitly-supervised factors:

- **L (Linguistic factor):** question marking, negation, topic marking, mouthing
- **A (Affective factor):** valence/arousal and discrete emotion

The factors are separated by training objectives (gradient reversal, feature orthogonality, mutual-information minimization) and the separation is *measured*, not asserted. The recovered affect then conditions two downstream generators:

1. An **emotion-conditioned text decoder** (so “I missed you” becomes “I’m so happy to see you again!” when the signer is joyful)
2. An **expressive avatar controller** (so the avatar’s blendshape intensity, motion amplitude and timing carry the emotion)

Key Innovation

The scientific core is objective O4: **Separating grammatical from affective non-manual signals**. This is achieved through:

- Gradient Reversal Layers (GRL) for adversarial factor separation
- Feature orthogonality penalties
- vCLUB (Variational Contrastive Log-ratio Upper Bound) mutual-information minimization
- Cross-prediction AUC measurement ($0.5 =$ perfect separation)

Technology Stack

Layer	Technology
Language	Python 3.11
Deep Learning	PyTorch 2.x
Landmark Extraction	MediaPipe Tasks API (HolisticLandmarker, FaceLandmarker)
Graph Neural Networks	ST-GCN family (decoupled spatial-temporal)
Sequence Models	GRU, SPOTER-style pose transformer
Translation	T5-small (google/t5-v1.1-small, 77M params)
FER Pretraining	DFEW, MAFW datasets
Smoothing	One Euro Filter
Experiment Tracking	Weights & Biases
Serving	FastAPI + WebSocket
Export/Runtime	ONNX, ONNX Runtime (CUDA EP), INT8 dynamic quant
Frontend	React + Three.js / React Three Fiber + three-vm
Avatar Format	VRM / Ready Player Me / Mixamo
Metrics	SacreBLEU, BERTScore, scikit-learn
Quality & Testing	pytest, ruff, pre-commit, mypy
Paper Writing	LaTeX (IEEEtran), BibTeX

Table 1: Technology Stack Overview

Deployment Target

- **Hardware:** 4 GB RTX 3050 GPU (laptop)
- **Latency:** End-to-end p95 < 400 ms
- **VRAM:** Peak < 2500 MB
- **Frame Rate:** ≥ 20 FPS capture

Estimated Timeline

Duration: 10 weeks (August 10 – October 18, 2026)

Phase	Duration	Key Deliverables
1. Foundations	Weeks 1–2	Repo, data pipelines, ASLLRP access, baselines
2. Recognition	Weeks 3–4	ST-GCN/SPOTER on ASL Citizen, keypoint pipeline
3. Affect	Weeks 5–7	Factorized encoder, GRL+vCLUB, EmoSign LOSO evaluation
4. Contribution	Weeks 8–9	Emotion-conditioned text, avatar controller, ablations
5. Ship	Week 10	Paper submission, code release, demo

Table 2: Project Phases

Critical Path

1. **Day 1:** Submit ASLLRP data-access request (gates core contribution O4)
2. **Weeks 1–2:** Establish data pipelines and verify disk space
3. **Weeks 3–4:** Achieve baseline recognition performance
4. **Weeks 5–7:** Core research — factorized non-manual encoder
5. **Week 10:** Paper submission and code release

Key Performance Indicators

KPI	Reference	Target
EmoSign emotion wF1 (video-only, LOSO)	GPT-4o: 20.76	≥ 35
Disentanglement cross-prediction AUC	0.5 = perfect	≤ 0.60
End-to-end p95 latency (RTX 3050)	—	< 400 ms
Peak inference VRAM	4096 MB limit	< 2500 MB

Table 3: Target KPIs

Datasets

Dataset	Content	Role in SEAM	Scale
ASL Citizen	Isolated signs, crowdsourced	Primary ISLR training	83,399 videos
EmoSign	Sentiment + emotions on ASLLRP	Primary affect benchmark	200 utterances
ASLLRP	Glosses + linguistic non-manuals	Source video + L supervision	2,651 utterances
How2Sign	Continuous ASL, keypoints	Phase B continuous SLT	~80 hours
INCLUDE	Indian SL, isolated	Cross-lingual ablation	4,287 videos

Table 4: Primary Datasets

Expected Outcomes

1. A $\sim 2\text{M}$ -parameter specialist model that separates grammatical from affective non-manual signals
2. State-of-the-art performance on the EmoSign benchmark (beating GPT-4o video-only)
3. End-to-end deployment on consumer-grade 4 GB GPU hardware
4. Emotion-conditioned text translation preserving semantic content
5. Expressive 3D avatar generation with emotion modulation
6. Peer-reviewed publication at a top-tier venue

Ethical Considerations

- Signer video is never redistributed (respects ASLLRP/ASL Citizen terms)
- Inference server binds to localhost only (no unauthenticated network exposure)
- Deaf-community involvement and limitations statement in the paper
- No claims of clinical, legal, or emergency-setting readiness
- Model cards for every released checkpoint with known failure modes